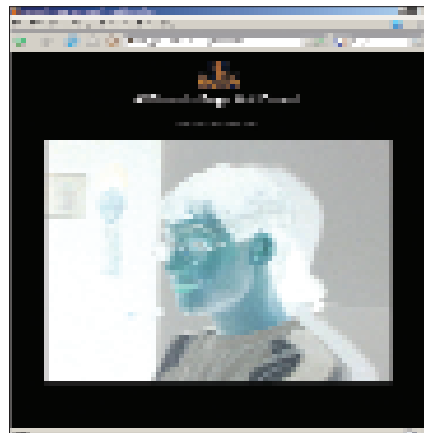


3DRealms Error 404 page
3DRealms is now best known for delaying *Duke Nukem Forever*...forever. Even they have a little Easter egg of their own. Enter a wrong URL to their site, say www.3drealms.com/blahblahblah. You will find a page with altered pic-



The people at 3drealms greet you when you type a wrong URL

tures of the people at 3drealms with random comments.

Winamp
Winamp has a challenging Easter egg in older versions of Winamp, which can make you lose your mouse click if you try hard. Get to Winamp's Preferences window by right clicking on Winamp, then on Options > Preferences. Go to Plug-ins > Input. Click on the Nullsoft Vorbis Decoder plugin, and click on the About button. A fish logo will appear on the left with the credits. Click on the fish and you will see it spin. Click a number of times, and a small meter measuring the rotations per minute will appear. Left-clicking on it rotates the fish to the left, and a right-click rotates it to the right. *Digit* is not



Head banging with the llamas

responsible for the loss or damage done to any mouse buttons!

Winamp has another trick that works with Winamp 5.0 and higher. First make sure you have the Modern skin selected and stretch the main window to the right using the right bottom edge till you find a Beat analyser at the centre. Hold down the [Ctrl] + [Alt] + [Shift] and click the centre till you get two llamas

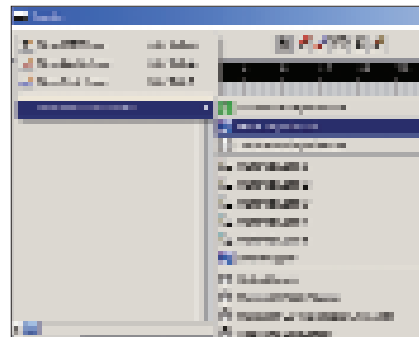
on either side. Now load up a song and play and watch the llamas' head bob in rhythm with the music.

FRINIKA

Professional audio sequencing software is costly, but they are also in plenty. For casual users, there are alternatives such as Frinika, an open source audio sequencer that runs on Java. Frinika (<http://frinika.sourceforge.net>) is available for Linux and Mac OS. The Windows version is on this month's DVD.

The Basics

Frinika will undoubtedly be a confusing experience to anyone who hasn't used a music sequencer before. When it starts the first thing to do, is select the output and input device. Choose to



Adding a synthesizer is the first step towards making a complete track

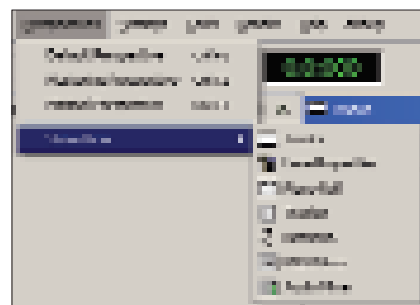
create a new project.

First, click on New > MIDI Out Device and select one of the synthesizers used to generate the sound. Next, select the instruments you want to use, by clicking on the MIDI Mixer tab. Set the volume for each instrument. Click on Null on the Lane Properties and choose the synthesiser that you created earlier. Now that the instruments are in place, you can use the Piano roll to draw bars on the chart. Click on the Write tool to add bars and right-click on the existing bars to delete them. To preview the tracks, click the Play button on the top of

the window. Additional effects can be added using the down arrow icon on the left bottom of the Piano Roll pane. Again, use the Write tool to drag and draw the pattern.

Changing The Type Of View

Frinika has a rather cluttered interface, with features and tool put into every empty crevice! To make things a little simpler, there are three different

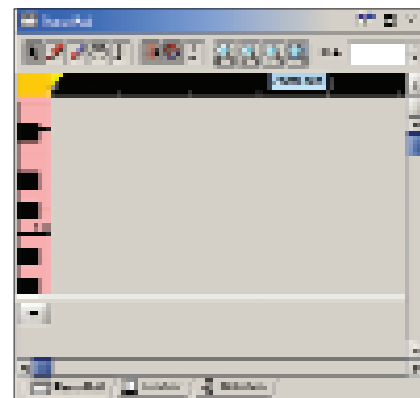


A choice of customized views are available for every task

presets of customised views present—Default, Master and Minimal. The master has all the features required, whereas the minimal perspective is to have ample working space while working on one feature of the program. These views can be accessed by clicking on the perspective menus or by the shortcuts [Ctrl] + [F1] / [F2] / [F3].

Performance Tweaks

Frinika runs on the Java environment, and along with other applications consume large amounts of memory. On Windows, Frinika runs as an executable,



By default, the Piano roll pane has very little empty workspace

but it can also be launched through command line, a batch file or a shortcut. We can then use parameters to increase or decrease the amount of memory consumed by Frinika. First, open Notepad and enter the following:

```
java -XX:ThreadPriority
Policy=1 -jar "c:\Program
Files\Frinika\frinika.jar".
Replace the c:\Program
Files\Frinika\frinika.jar with
the location of your Frinika installation. Go to File > Save and save the
file as "Frinika.bat" (use the quotes in the save dialog). Double-click this
batch file to run Frinika with those
parameters. The additional parameters
makes Windows run the application in
real-time as much as possible.
```

Phun

Phunny Physics

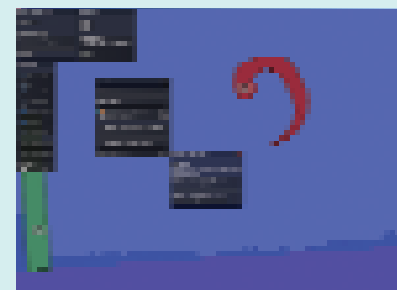
Rossi Fernandes

Phun! If you think we are not in our senses and that it's a weird way to say fun, then here goes—it's a free 2D physics simulation we gave out on the April '08 DVD. In short, you draw objects like you would on paper and all the drawings react with each other as though they are real. There's a lot to play around with, but where do you begin?

Understanding The Fundamental Operations

Before we start, here's a very basic introduction on how you handle objects and the workspace in Phun. You can create objects by clicking on the shapes in the toolbar on the left, and then clicking and dragging the scene. To draw your own custom shapes, choose the Brush tool and draw a shape but make sure you complete the loop. To rotate an object, right-click on the object and move it.

In typical empty scenes, there is no ground. Use the plane tool to draw a plane or all your objects will fall through empty space. If you want objects to be attached in the space, then click on the Fixate tool and click on the object created. When you have all the objects in place, you can press the Play button under Control. Everything now in the scene will react in real-time. To move objects around, you need to click the Drag tool. To navigate around the workspace, right-click and drag the scene. The mouse scroll can be used to zoom in and out.



Hinges can be set up to act as motors

Adding Hinge

Hinges allow you to fix objects and allow other objects to latch on to them and at the same time be

allowed some movement. Adding hinges in Phun can be done by simply creating the object that you want to fixate. Use the Hinge tool, and then click on a suitable location such as one end of the extremities of the object to allow a free movement. Click the Play button to have the object start rotating by gravity. The same hinge can also be converted to a motor. Right-click on the hinge and click on Hinges > Motor. Click the Motor checkbox to enable it. You can also choose the speed and power of the motor. To be able to manually control the motor, click on Control with arrow keys. Use the left and right arrow keys to control the speed of the motor.

Adding Springs And Chains

Springs are yet another component in Phun which allow two objects to interact. In springs, you can add tension between two objects just like any other spring. Similar to hinges, first create the object you to attach the strings to. Click the Spring tool and drag a spring from the blank space or one object to another. Spring properties such as strength and damping can be changed by right-clicking on the spring and then choose Spring in the menu that pops up.

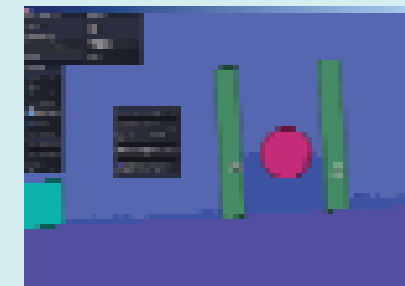
Chains are also attached in the same manner as springs by dragging lines between two points after selecting the Chain tool. Unlike springs, chains can be created without being attached to other objects. Chains can also be set up as motors like hinges by right-clicking on it and clicking on Chain again.

Creating And Grouping Objects

Keeping track of little objects can be a headache. If you're someone who is used to playing RTS games such as *Warcraft III*, then the same technique is used to create groups here. First, select the objects you want in your groups by drawing an enclosure using the brush tool. Press [Ctrl] + [number] to add all the selected objects to that group. To select the group, press that number again.

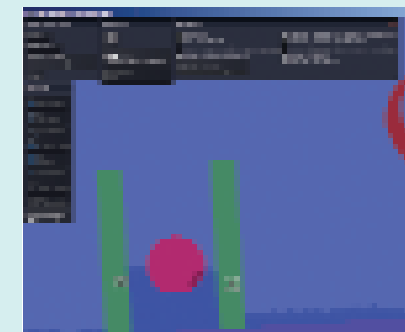
Liquid Simulation

Phun can do liquid simulations too. Liquids can't be created directly so you need to create shapes and then



Solid objects need to be converted to create liquids

convert them to liquids. So first draw the shapes like you normally would, then right click on the object and then click on Geometry settings > Liquify selected.



Some performance tweaks are necessary in CPU intensive simulations such as fluids

Tweaking Phun for Quality or Performance

Phun simulates everything in real-time so all the physics processing needs to be done in real-time by your processor. It's a heavy and strenuous task for the machine so a few settings can be altered when your scene becomes too complex to render smoothly. Click on Options under File and uncheck the box for Enable shaders and Draw clouds and check the one for Simple water rendering. If your machine handles Phun perfectly fine, then by all means leave shaders and clouds on and turn off simple water rendering.

If you're still left baffled by the things you can do with Phun, just load some ready made scenarios by clicking on File > Load Scene. You can then modify them or learn some new tricks from them. ■

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